Colloid Polym Sci (2014) 292:3347–3354
DOI 10.1007/s00396-014-3397-3

SHORT COMMUNICATION

Nanocomposites composed of P3HT:PCBM and nanoparticles synthesized by laser ablation of a bulk PbS target in liquid

N. G. Semaltianos · K. A. Maximova · A. I. Aristov · A. V. Kabashin · V. Heresanu · J.-Y. Hoarau · Z. Cao · L. Escoubas

Received: 17 April 2014 / Revised: 29 August 2014 / Accepted: 10 September 2014 / Published online: 21 September 2014
© Springer-Verlag Berlin Heidelberg 2014

Abstract Nanoparticles synthesized by laser ablation of bulk target materials in liquids have ligand-free surfaces since no chemical precursors are used for their synthesis, and thus, they are ideally suited for applications in the fabrication of organic solar cells in which the properties of the interface between the nanoparticles and the polymer blend matrix largerly determine the exciton splitting and transport of carriers to the external electrodes, properties crucial for the device operation and performance. Narrow band gap semiconducting quantum dots can act as sensitizers, increasing the absorption of the device active layer in the infrared part of the solar spectrum. In this work, a bulk PbS target was laser ablated (450 fs, 1,064 nm, 1 kHz) in ethanol for the synthesis of nanoparticle colloidal solutions. The solutions exhibit a broad absorption which extends at the longest wavelength measured of ~1,700 nm and beyond. The nanoparticles were directly mixed with the blend P3HT:PCBM for the formation of nanocomposites. The nanocomposites with the nanoparticles exhibit lower transmission in the whole spectral range as compared to the blend without the nanoparticles.

Keywords Laser ablation · Nanoparticles · Composites · PbS · P3HT:PCBM

PACS 79.20.Eb · 78.67.Bf · 85.80.Tc · 82.35.Np · 78.30.-j · 78.40.-q

Electronic supplementary material The online version of this article (doi:10.1007/s00396-014-3397-3) contains supplementary material, which is available to authorized users.

N. G. Semaltianos (✉)
Department of Physics, Aristotle University of Thessaloniki, Thessaloniki 54124, Greece
e-mail: nsemaltianos@yahoo.com

K. A. Maximova · A. I. Aristov · A. V. Kabashin
CNRS, LP3 UMR 7341, Aix Marseille Université, Campus de Luminy-Case 917, 13288 Marseille Cedex 9, France

V. Heresanu · J.-Y. Hoarau
CNRS, CINaM UMR 7325, Aix Marseille Université, Campus de Luminy-Case 913, 13288 Marseille Cedex 9, France

Z. Cao · L. Escoubas
Faculté des Sciences et Techniques, CNRS, IM2NP UMR7334, Aix Marseille Université et Sud Toulon-Var, Avenue Escadrille Normandie Niemen Service 231, 13397 Marseille Cedex 20, France

Introduction

In the field of organic electronics, nanocomposites consisting of polymer-fullerene blends with inorganic nanoparticles are very attractive for a use as active layers in organic solar cell (OSC) devices in the bulk heterojunction (BHJ) configuration since introduction of nanoparticles into the blend offers with the possibility of tuning the energy level and the absorption profile of the device towards the purpose of increasing its power conversion efficiency (PCE). The resulting device is termed as a ternary OSC [1]. In addition to II–VI based semiconductor quantum dots (SQDs) (CdSe, CdTe, ZnO) which exhibit absorption mainly in the visible part of the spectrum, IV–VI based (Pb chalcogenide) SQDs (PbS, PbSe, PbTe) are also preferred for this purpose, mainly because they are very efficient near infrared (NIR) quantum sensitizers due to their narrow band gaps. Investigations involving composites of the blend P3HT:PCBM with Pb-chalcogenide SQDs have been reported in the literature for PbS nanoparticles [2–4]. Incorporation of nanoparticles into the active layer of the device usually results in an increase of the absorption of the device.

One of the critical issues which determines the electrical properties of the blend-nanoparticle nanocomposite and thus the performance of the resulting device is related to the characteristic properties of the interface between the nanoparticles

and the polymer matrix which controls the exciton splitting (charge separation) and transport of the carriers to the external electrodes during device operation. The properties of the interface are largely determined by the presence of organic ligands on the nanoparticle surfaces. Semiconducting nanoparticles synthesized via chemical methods utilizing organometallic chemical precursors [5, 6] bear on their surfaces insulating organic ligands and passivation layers such as oleic acid, oleyalamine, chloride or trioctylphosphine oxide (TOPO) which on one side renders them soluble in common organic solvents, a property important for the process of the nanoparticles from solution but on the other side are detrimental for the device operation. Synthesis of PbS nanoparticles via solid-state reactions based on non-ionic surfactant-based microemulsions also lead to the appearance of a shell of the surfactant surrounding the nanoparticles [7]. These ligands should be removed and exchanged with other electrically conducting ligands such as pyridine, benzenedithiol, alkylamines or others [8] before the nanoparticles are introduced into the blend, procedures which usually involve time-consuming and solvent-wasting steps. In situ chemical synthesis of the nanoparticles within the polymer matrix [9] also results in the presence in the composite of reaction by-products which act as contaminants. Other methods for the fabrication of PbS nanoparticle-polymer composites such as cluster beam deposition of the nanoparticles simultaneously with the organic material onto substrates [10], althought they produce nanoparticles with ligand-free surfaces, however, do not allow the deposition of the composites from solution and might not be compatible with the objective of fabricating devices on flexible substrates in the roll-to-roll manufacturing technology.

A well-established alternative method for the synthesis of nanoparticles involves laser ablation of a bulk target material while the material is immersed in a liquid [11]. The nanoparticles are formed by the debris which is generated as a result of the material's ablation, and a nanoparticle colloidal solution is formed. Since no chemical precursors are utilized for the nanoparticle synthesis, colloidal solutions with reduced reaction by-products, and most importantly nanoparticles with ligand-free (bare) surfaces, are produced. This particular property makes semiconducting nanoparticles synthesized by laser ablation of bulk target materials in liquids attractive for their utilization in the fabrication of OSCs.

In this work, a bulk PbS target was laser ablated in ethanol for the synthesis of nanoparticles. The nanoparticles were subsequently mixed with the blend P3HT:PCBM for the formation of nanocomposites and the optical and morphological properties of the nanocomposite were investigated for its potential use as the active layer in OSC devices.

Experimental details

The bulk target material was a piece of PbS (galena) semiconductor (E_g=0.37 eV) with dimensions of 10×5×5-mm cut from the single crystal rod which was grown by the vertical Bridgman growth method starting from the elemental powders [12]. It was ablated using a femtosecond pulse laser (450 fs, 1,025 nm, 1 kHz, 96 or 20 μJ) while it was immersed in absolute ethanol in a small beaker with 10-mm liquid level above its surface. The beam was impinging vertically onto the horizontally lying target at the bottom of the beaker, scanning the target surface in a meander fashion to expose a fresh area for ablation. Ablation was performed for 5 mins (1 ablation cycle) yielding 0.6 mg of nanoparticles in the 5-ml volume of the liquid (concentration 0.12 mg/ml) determined by weighting the target before and after ablation. No bubbles were observed to adhere onto the sample surface during ablation which would otherwise had shielded or scattered the laser radiation. The resulting colloidal solutions were stable against sedimentation for 1 day. Nanoparticle samples for characterization were prepared by drying out the colloidal solutions onto glass substrates as well as on carbon-coated copper grids for transmission electron microscopy (TEM) characterization.

In order to prepare enough quantity of nanoparticles to be mixed with the polymer-fullerene blend, ablation was repeated for the required number of cycles to obtain the desired mass of nanoparticles, the colloidal solutions were collected in the same container and enough liquid volume was evaporated from the final solution to achieve the desired nanoparticle solution concentration. The nanocomposite of the polymer-fullerene blend with the nanoparticles (P3HT:PCBM:NPs) was in a weight ratio of 1:0.9:0.25. First, the polymer-fullerene blend was prepared by dissolving P3HT (≥95 % regioregular, Rieke Metals) and PCBM (C_{70} PCBM 99 %, Nano-C) in 1,2-dichlorobenzene and stirred overnight at 75 °C. Then, the nanoparticle colloidal solution (immediately after formation) was mixed with the polymer-fullerene blend under stirring for ~40 mins. The resulting nanocomposites were deposited by spin coating (2,600 rpm for 80 s) onto clean glass substrates followed by baking on a hot plate at 160 °C for 10 mins inside a nitrogen-filled glovebox. X-ray diffraction (XRD) of the nanoparticles was performed by using a diffractometer with a Cu K_α source (λ=1.5406 Å) and TEM/HRTEM by using a JEOL JEM-3010 instrument. Atomic force microscopy (AFM) of the nanocomposites was performed in non-contact mode using Si cantilevers with a given radius of curvature less than 10 nm. Raman microspectroscopy was performed using excitation at 532 nm. The nanoparticles were also characterized by X-ray photoelectron spectroscopy (XPS) using a spectrometer with Mg Kα primary radiation (1253.6 eV).

Results and discussion.

Synthesis and characterization of nanoparticles by laser ablation of the bulk PbS target material in ethanol

During the ultrashort pulse laser ablation, the laser pulse with energy well above the band gap of the material is incident onto the material surface and is absorbed via linear absorption. Electrons are excited to the conduction band leaving holes behind in the valence band, and a highly energetic and dense carrier population is created forming an electron-hole plasma (EHP). The plasma transfers its energy to the lattice via electron-phonon coupling which leads to the destabilization of the lattice and finally to the ablation of the material via the formation of a plasma plume of ablated species corresponding to sublimation of the material. The nanoparticles are formed by nucleation of the ablation plume species in the vapour phase during the expansion and adiabatic cooling of the plume within the liquid. However, for high laser pulse energies much above the ablation threshold of the material, there is diffusion of the absorbed laser energy in the form of heat outside of the volume in which the radiation is initially absorbed and a percentage of the formed nanoparticles correspond to material which is ablated in the form of melted droplets (via melting and vaporization).

XRD patterns measured from the nanoparticles are shown in Fig. 1 together with the pattern of the bulk material. In addition to the peaks corresponding to the rock-salt (cubic) crystal structure of PbS (JPCDS No. 5–592), it is readily seen that a few other peaks are also present in the patterns from the nanoparticles. These peaks could be assigned to reflections from the orthorhombic crystal structure of lead sulphate (anglesite) $PbSO_4$ (JPCDS No. 36–1461). This indicates that the synthesized nanoparticles are a mixture of PbS and $PbSO_4$. $PbSO_4$ is formed from the oxidation of PbS by the reaction with oxygen: $PbS+2O_2 \rightarrow PbSO_4$. The oxygen atoms which are needed for the interaction with the PbS nanoparticles originate from the thermal decomposition of ethanol at the plasma plume-liquid interface induced by the high temperature of the expanding plasma plume. It is known that thermal decomposition (pyrolysis) of ethanol via C–C or C–O bond dissociation produces among others, oxygen radicals, and these reactions of ethanol decomposition become favourable mainly at very high temperatures (>2,500 K) [13, 14]. Thus, such a reaction can become dominant under the high-temperature and high-pressure conditions of the laser plasma plume. Oxidation of bulk PbS in oxygen atmosphere takes place at temperatures below 850 °C [15]. However, oxidation of PbS nanoparticles to $PbSO_4$ was observed to take place by heating the nanoparticles in air at 250 °C while heating at above that temperature lead to the formation of lead oxysulphate [16]. Since oxysulphates are not observed at the present experiments, it is likely that oxidation of the PbS nanoparticles to $PbSO_4$ takes place towards the latest stages of plume expansion and in particular in the region close to the plume-liquid interface when the temperature of the plume has become lower than 250 °C because of the adiabatic cooling just before quenching and possibly after the nucleation and formation of PbS nanoparticles. In the pattern from the nanoparticles synthesized by using a higher pulse energy of 96 μJ, the intensities of the peaks corresponding to PbS and $PbSO_4$ increase by the same percentage for both materials as compared to the intensities of the peaks for the two materials in the pattern from the nanoparticles synthesized using a lower pulse energy of 20 μJ. This indicates that the relative mass proportion of the two materials in a certain nanoparticle ensemble is independent of the pulse energy—using a higher pulse energy only leads to the synthesis of a larger total mass of nanoparticles (solution with a higher nanoparticles concentration).

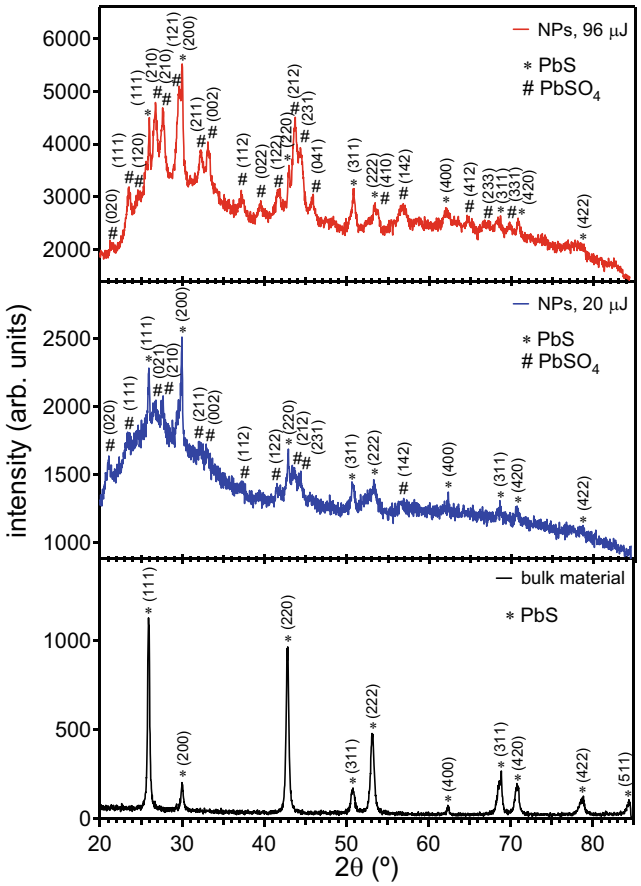

Fig. 1 XRD patterns of the nanoparticles

TEM images of the nanoparticles and size distribution histograms are shown in Fig. 2a–d, respectively. HRTEM images and electron diffraction patterns are shown in Figs. S1 and S2. It is seen that the nanoparticles are mostly spherical for both energies. Nanoparticles with diameters from 2 and up to 22 or 26 nm are obtained for the two pulse energies, respectively. Their size distribution histograms by counting approximately 500 particles in images of

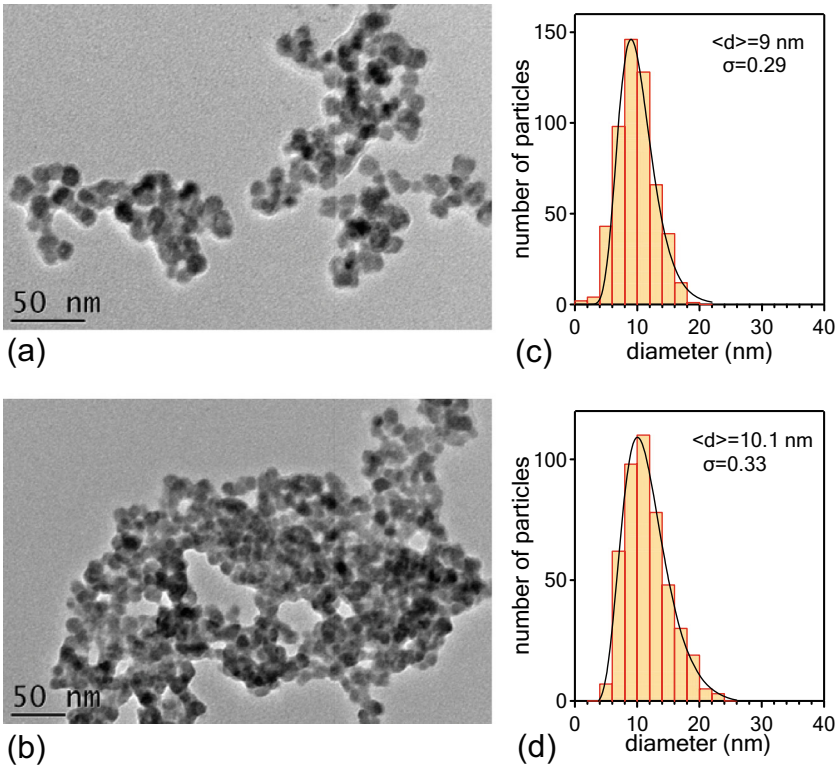

Fig. 2 TEM images and size distribution histograms of the nanoparticles synthesized using pulse energy of 96 μJ (**a**, **c**) and 20 μJ (**b**, **d**)

nanoparticle ensembles obtained on different areas on each grid are shown in Fig. 2c, d). The histograms are described quite well by log-normal functions with median diameters of $<d_0>=9$ and 10.1 nm and standard deviations of $\sigma=0.29$ and 0.33, respectively. The slightly larger median diameter and broader size distribution of the nanoparticles synthesized using the lower pulse energy might be due to the weaker fragmentation effect of the synthesized nanopartcles which takes place by the ablating laser beam during their synthesis, due to the lower pulse energy.

Raman spectra from the nanoparticles are shown in Fig. 3. The spectra are characterized by a strong band at 978 cm^{-1}

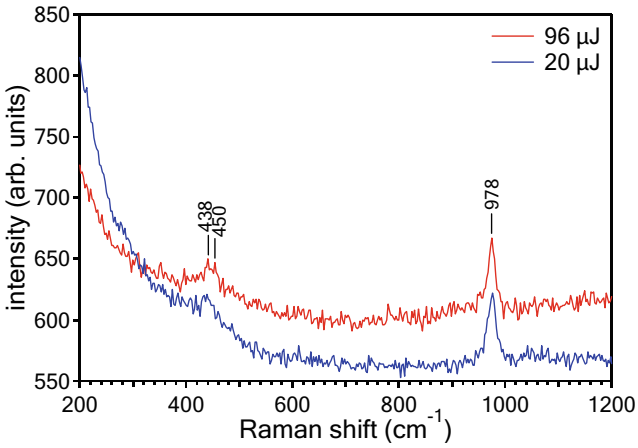

Fig. 3 Raman spectra of the nanoparticles (excitation energy of 0.35 mW, exposure time of 10 s)

and a split band at low wavenumbers consisting of two lines at 438 and 450 cm^{-1}. This spectrum is attributed to $PbSO_4$ [17] which confirms once again the presence of anglesite in the synthesized nanoparticles in accordance with the XRD patterns (Fig. 1). The Raman spectrum from bulk PbS is characterized only by a weak band at 451 cm^{-1} (see Fig. S3 of the Supporting Information), but this band could not be distinguished in the spectra from the nanoparticles due to the appearance in the same wavenumber region of the band from the $PbSO_4$.

The absorption spectrum of the nanoparticle colloidal solution immediately after synthesis is shown in Fig. 4 determined by measuring their transmission spectrum using transmittance and converting it to absorbance using the relation $A=-\ln(T)$ [18]. In this case of a nanoparticle colloidal solution in which the nanoparticles have diameters much smaller than the wavelength of light, there is almost one-to-one correspondence between the measured transmittance and the absorbance calculated from the above relation. In the same graph, the measured transmission spectrum of the nanoparticles on the glass substrate (after drying out the colloidal solution on the substrate) is shown. A broad absorption is obtained in the whole spectral range which starts at wavelengths even higher than the measured maximum wavelength of 1,687 nm—inferred by the fact that the slope of the absorption curve even at the maximum measured wavelength is not 0. Monodisperse PbS nanoparticles with a diameter of 9.5 nm (E_g=0.75 eV) synthesized by chemical methods exhibit a

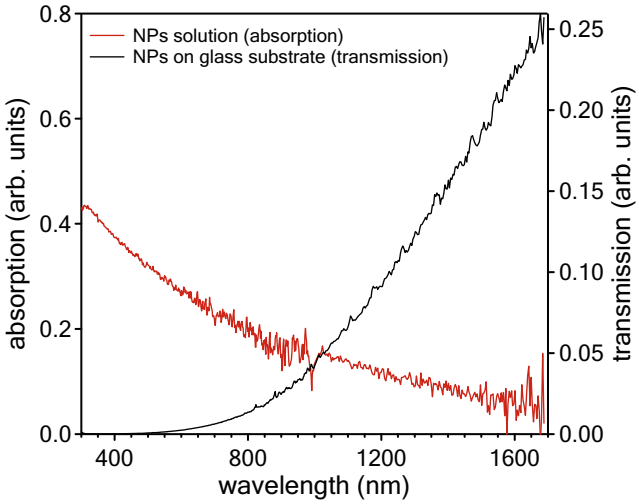

Fig. 4 Absorption and transmission spectra of the nanoparticles in solution and of the solution dried onto a glass substrate, respectively

narrow band gap absorption peak at 1,653 nm according to the quantum confinement model for quantum dots [19] while for nanoparticles with diameter >15 nm, the band gap absorption is observed at wavelengths >2,000 nm. However, although close to the maximum wavelength which can be measured by the spectrometer, an absorption peak from the present nanoparticles is not observed due to their relatively wide size distribution (the spikes in the spectrum are noise due to the weak signal). Each nanoparticle contributes to the total absorption a small percentage of absorption but at a different wavelength due to their variety of sizes. The number of nanoparticles with the same diameter is not large enough for their contribution to appear as a peak in the total absorption curve. The fact that the nanoparticles are not pure PbS but a mixture of PbS and $PbSO_4$ also contributes to this fact. Thus, the absorption curves which are obtained at the present case are consistent with what is expected to be obtained from the quantum confinement model for quantum dots for nanoparticles with diameters from 2 and up to 22–26 nm.

XPS spectra from the nanoparticles on the glass substrate are shown in Fig. 5. All spectra are referenced to the C 1 s line at 285 eV. In the Pb 4f region (131–150 eV) (Fig. 5a), the peaks at 137.9 and 142.8 eV correspond to Pb $4f_{7/2}$ and Pb $4f_{5/2}$, respectively [20] while the peak at the low-energy side of the Pb $4f_{7/2}$ peak (at ~134.15 eV) is due to the satellite peaks from the Mg Kα source. In the S 2p region (155–175 eV) (Fig. 5b), the peak at 159.45 eV is due to sulphide (S^{2-}) bound to Pb^{2+}, the peak at 163.3 eV is due to S bound to one, two or three oxygen atoms (SO_n) and the peak at 168.65 eV is due to SO_4 [21].

Nanocomposite films of P3HT:PCBM with nanoparticles

In the optical microscopy image of the nanocomposite film (blend with nanoparticles) (Fig. 6b), dark areas of material accumulation are seen which are attributed to nanoparticle agglomerates since such dark areas are not observed in the blend without the nanoparticles (Fig. 6a). These agglomerates lie within the volume of the nanocomposite, and they have also been observed in nanocomposites involving nanoparticles synthesized by chemical methods [22, 23]. Formation of nanoparticle agglomerates is particularly attributed to the fact that the nanoparticles have surfactant-free surfaces and it is likely that prolonging the time duration of the mixing of the nanoparticles with the blend and/or using ultrasonication together with light heating will prevent the formation of large aggregates [24].

AFM image of the blend (without the nanoparticles) (Fig. 6c) exhibits the typical surface morphology of P3HT:PCBM blend after thermal annealing, consisting of narrow interconnected islands separated by narrow deep trenches [25]. The PCBM particles are distinguished as small spherical areas with dimensions in the order of 50 nm (marked with the circles in the image). In the AFM image of the surface of the nanocomposite film (Fig. 6d), the nanoparticle aggregates which are partially projecting out of the film surface are easily distinguished as relatively large and irregular shape areas (dimensions of the order of 200–500 nm, marked in the image). The root mean square roughness of the film increases from 3.31 to 15.69 nm.

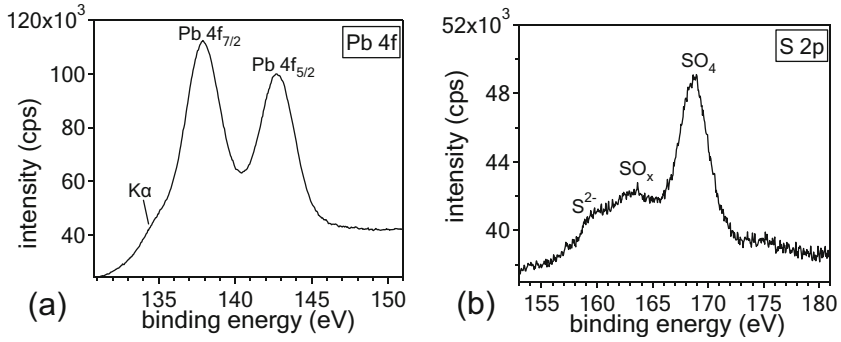

Fig. 5 XPS spectra of the nanoparticles

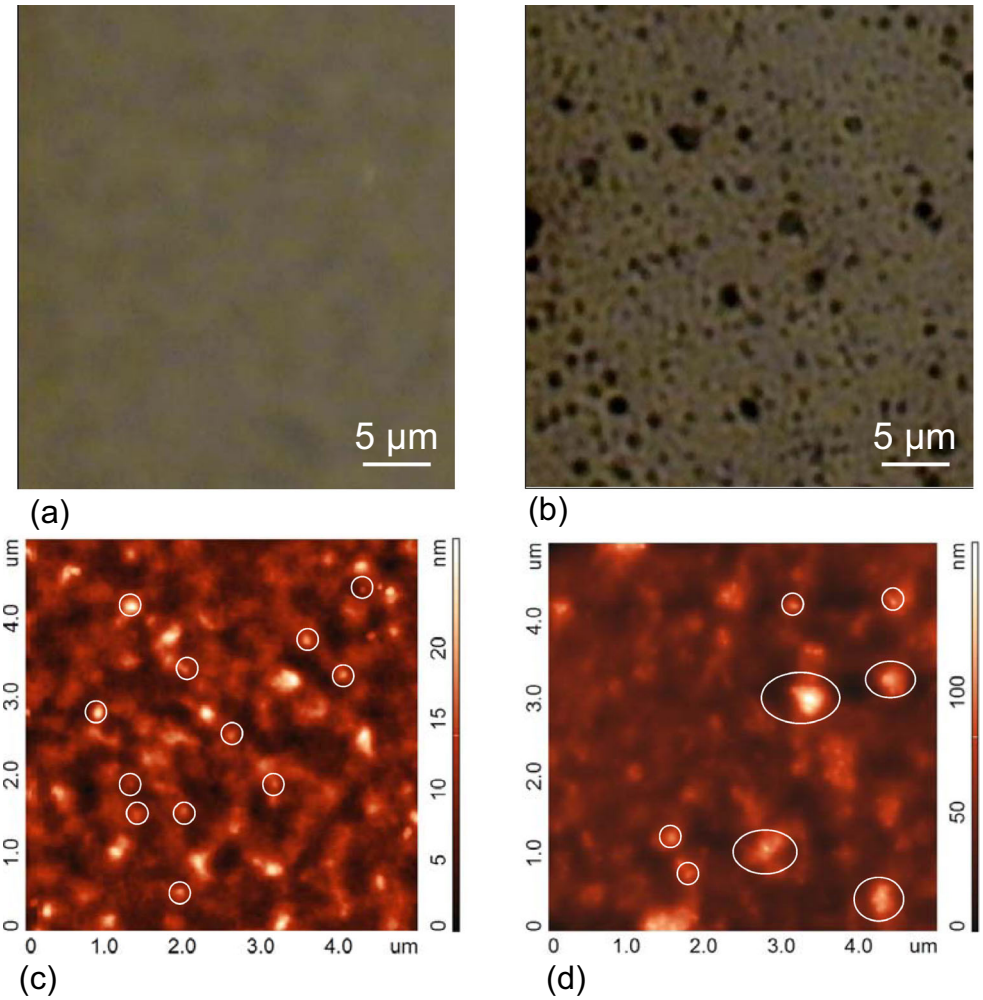

Fig. 6 Optical microscopy and AFM images of the film without the nanoparticles (**a**, **c**) and with the nanoparticles (**b**, **d**), respectively

In the Raman spectrum from the nanocomposite films (Fig. 7), the peak from the $PbSO_4$ of the nanoparticles appears at ~980 cm^{-1} as a shoulder in the left hand side of the C_β–C_{alkyl} stretching band of the P3HT at 998 cm^{-1}. Other peaks in the spectra are identified in Table 1 based on reports in the literature [26]. The strongest peak at 1,463 cm^{-1} ("pentagonal pinch"), which characterizes the Raman spectra of PCBM [27], cannot be distinguished in the spectra due to the overlap with the C_α=C_β stretching band from the P3HT at 1,446 cm^{-1}. It is further observed that the FWHM of the C_α=C_β stretching

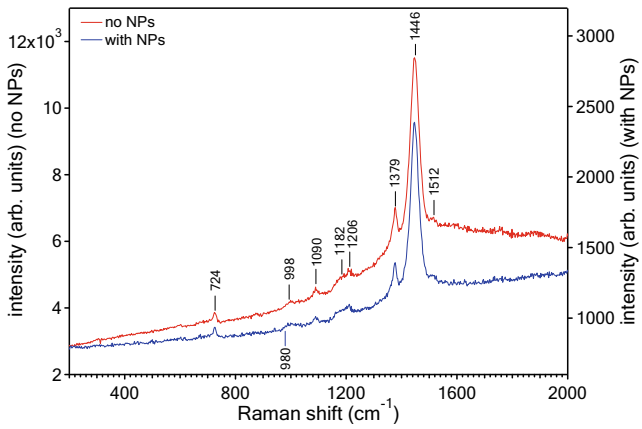

Fig. 7 Raman spectra of the films without and with the nanoparticles (excitation energy of 1 mW, exposure time of 10 s)

Table 1 Vibrational modes in the Raman spectra of P3HT:PCBM-PbS/$PbSO_4$ NPs nanocomposite films.

Vibrational mode frequency (cm^{-1})	Assignment
1,512	$C_{\alpha'}$=$C_{\beta'}$ stretching (P3HT)
1,446	C_α=C_β symmetric stretching (″)
1,379	C_β–$C_{\beta'}$ intra-ring stretching (″)
1,206	C_α–$C_{\alpha'}$ inter-ring stretching (″)
1,182	C_α–$C_{\alpha'}$ inter-ring stretching (″)
1,090	C_β^+–H bending (″)
998	C_β–C_{alkyl} stretching (″)
724	C_α–S–$C_{\alpha'}$ deformation (″)
980	$PbSO_4$ of NPs

band (~34 cm^{-1}) as well as the intensity ratio of the C_β–$C_{\beta'}$ intraring stretching band with this band (~0.12) is the same for both films, and this indicates that incorporation of the nanoparticles into the blend does not disturb the molecular order of the polymer [28].

In the graph of Fig. 8, measured transmission spectra of the films are also shown, and they are consistent with the typical spectra of annealed blend P3HT:PCBM films which have been reported in the literature [28, 29] considering that transitions which in the absorption spectra show as peaks, in the transmission spectra appear as deeps, instead. In particular, transmission starts decreasing at ~635 nm which corresponds to the optical gap of P3HT. The deep at 602.43 nm is due to a singlet excitonic transition with vibronic (phonon) side bands at 543.85 and 513.75 nm. The deep at 470 nm is due to interband transition and corresponds to the fundamental electronic gap of the P3HT:PCBM blend [30]. The deep at 334.73 nm is due to the $S_0 \rightarrow S_{17}$ electronic transition of PCBM [31]. More importantly, it is seen that the shape of the deep at 602.43 nm and of the overall structure of the transmission spectrum in general is preserved in the nanocomposite film, and this indicates that incorporation of the nanoparticles into the blend does not disrupt the molecular packing of the polymer [28]. The nanocomposite film exhibits lower optical transmission by ~6 % as compared to the blend without the nanoparticles, with ~80 % of this amount in the visible region from 300 to 700 nm.

The refractive index and extinction coefficient of the films are shown in Fig. 9 determined from variable angle spectroscopic ellipsometry in combination with transmission measurements. Data of the ellipsometric angles Ψ and Δ together with the theoretical fittings are shown in Fig. S5. For the fitting, a simple B-spline model was considered with surface roughness and allowing for Kramers-Kronig consistency. It is known that in the case of annealed P3HT:PCBM films, a better fitting of the ellipsometric data is obtained considering an anisotropic model [32]; however, at the present analysis for simplicity, we considered an isotropic model. For the nanocomposite with nanoparticles, the Bruggeman effective medium approximation (EMA) was considered assuming that the nanoparticles are pure PbS and allowing their percentage composition as a fitting parameter in the model. The resulting thicknesses of the films were determined as 92.57±0.056 and 106.90±0.45 nm for the film without and with the nanoparticles, respectively. From the results presented in Fig. 9, it seems that the extinction coefficient for the nanocomposite film is lower in the whole spectral region measured; however, as it is well known, ellipsometry measurements of films which exhibit considerable surface roughness might be meaningless due to the large percentage of radiation which is randomly scattered from the film surface and not reaching the detector; thus, the results of Fig. 9 should be taken with caution.

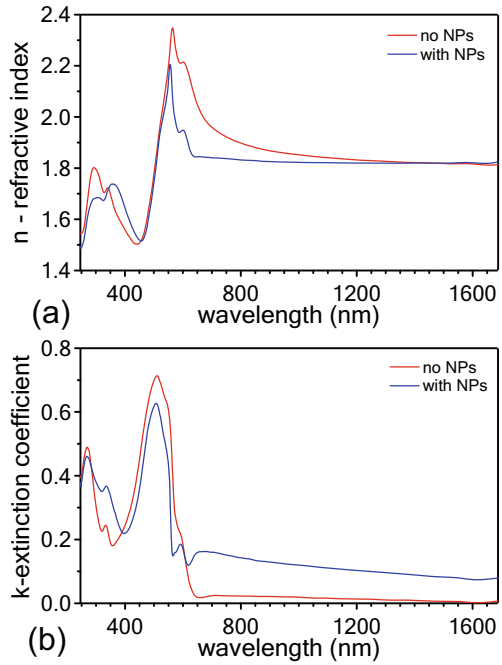

Fig. 9 Refractive index (**a**) and extinction coefficient (**b**) of the films determined from spectroscopic ellipsometry measurements

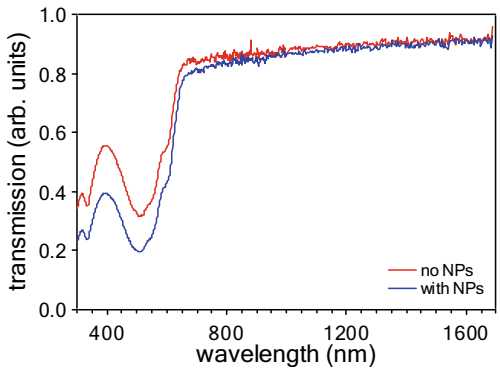

Fig. 8 Transmission spectra of the films with and without the nanoparticles

Conclusions

In this paper, the potential of using nanoparticles synthesized by laser ablation of a bulk PbS target in liquid in the fabrication of the polymer-fullerene/nanoparticles nanocomposite of the active layer of OSC devices was investigated. The nanoparticles synthesized in ethanol are a mixture of PbS and $PbSO_4$ due to the oxidation of the synthesized PbS nanoparticles by the oxygen from the thermal decomposition of the ethanol at the plasma plume-liquid interface induced by the high temperature of the expanding plasma plume. The nanoparticle colloidal solution exhibits a broad absorption extending in the

infrared region of the spectrum up to the maximum measured wavelength of ~1,700 nm. Nanocomposite thin films spin coated onto glass substrates fabricated by mixing the synthesized nanoparticles with the P3HT:PCBM blend exhibit transmission in the whole spectral region lower than the blend films without the nanoparticles.

Acknowledgments The research leading to these results has received funding from LASERLAB-EUROPE (grant agreement no. 284464, EC's Seventh Framework Programme), project CNRS-LP3001943 "Nanoparticles by Laser Ablation and Composites".

References

1. Chen Y-C, Hsu C-Y, Lin RYY, Ho K-C, Lin JT (2013) ChemSusChem 6:20
2. Itskos G, Othonos A, Rauch T, Tedde SF, Hayden O, Kovalenko MV, Heiss W, Choulis SA (2011) Adv Energ Mater 1:802
3. Rauch T, Böberl M, Tedde SF, Fürst J, Kovalenko MV, Hesser G, Lemmer U, Heiss W, Hayden O (2009) Nat Photon 3:332
4. Nam M, Kim S, Kang M, Kim S-W, Lee K-K (2012) Org Electr 13: 1546
5. Hines MA, Scholes GD (2003) Adv Mater 15:1844
6. Murray CB, Norris DJ, Bawendi MG (1993) J Am Chem Soc 115: 8706
7. Wang W, Liu Y, Zhan Y, Zheng C, Wang G (2001) Mater Res Soc Bull 36:1977
8. Fridaus Y, Khetubol A, Kudret S, Dilien H, Maes W, Lutsen L, Vanderzande D, M. Van der Auweraer (2012) Photon. Sol. Energ. Syst. IV, SPIE Proc. 8438
9. Watt A, Rubinsztein-Dunlop H, Meredith P (2005) Mat Lett 59:3033
10. Zachary AM, Bolotin IL, Asunskis DJ, Wroble AT, Hanley L (2009) ACS Appl Mater Interf 1:1770
11. Yang GW (2007) Prog Mater Scie 52:648
12. Gutknecht K-H, Preier H, Hesse J (1975) Mater Res Soc Bull 10:277
13. Li J, Kazakov A, Dryer FL (2004) J Phys Chem A 108:7671
14. Yang S, Cai W, Zeng H, Xu X (2009) J Mater Chem 19:7119
15. Kamchatka MI, Chashchinov YM, Chesnokova DB (2001) Inorg Mater 37:910
16. Mozafari M, Moztarzadeh F, Vashaee D, Tayebi L (2012) Phys E 44: 1429
17. Batonneau Y, Brémard C, Laureyns J, Merlin JC (2000) J Raman Spectr 31:1113
18. Ng A, Li CH, Fung MK, Djurišić AB, Zapien JA, Chan WK, Cheung KY, Wong W-Y (2010) J Phys Chem C 114:15094
19. Wang Y, Suna A, Mahler W, Kasowski R (1987) J Chem Phys 87: 7315
20. Zlngg DS, Hercules DM (1978) J Phys Chem 82:1992
21. Yashina LV, Zyubin AS, Püttner R, Zyubina TS, Neudachina VS, Stojanov P, Riley J, Dedyulin SN, Brzhezinskaya MM, Shtanov VI (2011) Surf Scie 605:473
22. Verma D, Dutta V (2009) J Renew Sustain Ener 1:023107
23. Khan MT, Kaur A, Dhawan SK, Chand S (2011) J Appl Phys 110: 044509
24. Nguyen VS, Rouxel D, Vincent B (2014) Ultrason Sonochem 21:149
25. Reyes MR, Kim K, Carroll DL (2005) Appl Phys Lett 87:083506
26. Trznadel M, Zagorska M, Lapkowski M, Louarn G, Lefrant S, Pron A (1996) J Chem Soc, Faraday Trans 92:1387
27. Yoo SH, Kum JM, Cho SO (2011) Nanosc Res Lett 6:545
28. Tsoi WC, James DT, Kim JS, Nicholson PG, Murphy CE, Bradley DDC, Nelson J, Kim J-S (2011) J Am Chem Soc 133:9834
29. Lioudakis E, Othonos A, Alexandrou I, Hayashi Y (2007) Appl Phys Lett 91:111117
30. Sakurai K, Tachibana H, Shiga N, Terakura C, Matsumoto M, Tokura Y (1997) Phys Rev B 56:9552
31. Chirvase D, Parisi J, Hummelen JC, Dyakonov V (2004) Nanotechn 15:1317
32. Ng A, Liu X, To CH, Djurišić AB, Zapien JA, Chan WK (2013) ACS Appl Mater Interf 5:4247